

Dayoung Kang

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KEYWORDS

Machine Learning, Deep Learning, Multimodal Learning, Representation Learning, AI4Science, AI Agent, Large Language Models, Microsoft Azure, Python, Linux

EDUCATION

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

M.S., Electrical Engineering and Computer Science

Daegu, Korea

Feb 2025 – Current

- Advisor: Prof. Jinhyun So

Jeonbuk National University

B.S., Double Major

Jeonju, Korea

May 2021 – Feb 2024

- Division of Computer Science and Engineering
- Department of Smart Farm
- Cumulative GPA: 4.14/4.50; Major GPA: 4.21/4.50

EXPERIENCE

Graduate Researcher

DGIST

Collaborative, Reliable, and Efficient AI Lab., Advisor: Prof. Jinhyun So

Feb 2025 – Current

- Developed a unified olfaction framework for single molecules and two-molecule mixtures (ICASSP 2026)
- Research Interests: multimodal learning and agentic AI

Undergraduate Researcher

Jeonbuk National University

Medical AI & Computational Science Lab., Advisor: Prof. Kyungsu Lee

Apr 2024 – Dec 2024

- Studied rotator cuff segmentation from ultrasound images

Undergraduate Researcher

Jeonbuk National University

Smart Digital Agriculture Lab., Advisor: Prof. Taegon Kim

Feb 2022 – Nov 2022

- Modeled and analyzed crop yields for paprika and wheat using tabular growth data

PUBLICATIONS

Google Scholar: https://scholar.google.com/citations?user=UX_2oycAAAAJ&hl

D.Kang, JW.Kim, J.Park, K.Lee, JW.Choi, and J.So. “AROMMA: Unifying Olfactory Embeddings for Single Molecules and Mixtures”, in *2026 International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2026.

PROJECTS

Bidirectional Olfactory Converter Agent

Aug 2025 – Oct 2025

- LLM-powered odor prediction and recommendation agent leveraging AROMMA (ICASSP 2026)
- Tech Stack: RAG (Retrieval-Augmented Generation), LangChain, Prompt Engineering
- Link: <https://github.com/riverallzero/aromma-agent>

AWARDS

President's Award, Jeonbuk National University

2024

Grand Prize (Entrepreneurship Award), San Jose State University

2024

- Proposed RENNIS (Robot+Tennis), an personal tennis coaching robot

Best Excellence Award (K-Health Medical AI Hackathon-Track 01), Korea Smart Healthcare Association

2024

- Developed breast mass segmentation model from mammography images
- Link: <https://github.com/riverallzero/maithon>

Special Award in Service Category (ChatGPT Microsoft AzureThon), AI Factory

2023

- Developed chatbot that converts business card images into contact files by integrating multiple APIs
- Link: <https://github.com/riverallzero/azurethon>